ANDREA DI FRANCIA

Senior Data Scientist | Machine Learning Engineer | Health Economics

andrea.l.difrancia@gmail.com   andreadifra   andreadifrancia   London, UK

SUMMARY

Senior Data Scientist and Machine Learning Engineer with 8+ years of experience delivering NLP, forecasting, Bayesian, survival, and health-economics models across healthcare and government. Skilled in Python, R, SQL, Snowflake, AWS, Azure, Docker, Airflow, and MLflow, building production ML systems, decision tools, and quantitative evidence for clinical, public-health, and policy stakeholders.

EXPERIENCE

Data/Machine Learning Scientist (Contractor)

Bupa

Nov 2025 – Present — Remote, UK

- Restructured **Snowflake**-based health assessment pipelines, enabling longitudinal analysis across **107,000+ patient records** and producing core evidence for an academic white paper on customer health outcomes.
- Implemented prototype **QRISK3/QDiabetes** risk models and a **Markov cost-effectiveness model** for Bupa's AI personalisation strategy, with preliminary results indicating **£1,700 savings per individual** and **0.30 QALYs gained**.
- Authored the production roadmap for AI-powered **Personalised Health Plans**, defining **MLOps architecture**, clinical governance, and **Azure/Snowflake** deployment for integration into the Blua patient app.

Lead Data Science Consultant

Health Economics and Outcomes Research (HEOR)

Oct 2024 – Nov 2025 — Remote, UK

- Built **zero-inflated negative binomial** and **mixture regression models** on linked healthcare data to predict 3-year obesity-related costs, quantifying **£3,000+ savings per patient** from reducing type 2 diabetes risk.
- Analysed **161,000+ T2D patients** using **parametric survival models**, showing materially higher 10-year CVD and kidney risk for high-BMI cohorts and informing targeted screening strategies.
- Led a global meta-analysis of meningococcal carriage using **hierarchical mixed-effects logistic regression**, producing age- and region-specific prevalence estimates to support vaccination strategy.

Senior Data Scientist

UK Health Security Agency

Jan 2022 – Sep 2024 — London, UK

- Developed and deployed **transformer-based NLP classifiers** extracting **8 hepatitis serology markers** from around **5,000 clinical reports/month**, achieving **95%+ performance** and saving **10 hours/month** of specialist review.
- Built a **Bayesian model** in **R/Stan** with **University of Oxford** collaborators, estimating the NHS Covid-19 App prevented roughly **1.4m cases**, **40k hospitalisations**, and **3k deaths** during Omicron.
- Engineered a **Python ETL pipeline** for avian-influenza surveillance, automating ingestion from tabular and PDF sources, adding geographic enrichment, and cutting manual processing by **5 hours/week**.
- Designed **non-linear regression models** on genomic sequencing data to estimate Omicron sub-lineage prevalence and support epidemiological surveillance.
- Productionised public-health analytics on **AWS** and **Azure** using **Docker**, **Airflow**, **MLflow**, and scheduled inference workflows, while line-managing junior data scientists in a cross-functional delivery team.

Economic Data Scientist

Department for Education

Apr 2017 – Dec 2021 — London, UK

- Developed **R**-based forecasting models for the **£24bn teacher pay-bill** and deployed **Shiny** scenario dashboards used by finance and policy teams across the department.
- Quantified the economic case for a successful **£240m Early Career Framework** proposal and built rollout-monitoring **Shiny** dashboards that received an award for outstanding achievement.
- Contributed to the **Schools' Cost Pressures Model** forecasting around **£40bn** of expenditure and upgraded the **Apprenticeship Levy Model** to cover **3,000 additional schools** and about **£100m** of levy spending.
- Modelled teacher supply elasticity for the **School Teachers' Review Body** and led higher-education financial-sustainability analysis, informing policy decisions on workforce incentives and government intervention.

EDUCATION

MSc. Applied Statistics & Operational Research

Birkbeck, University of London

Distinction

- Top 10% in cohort; highest marks: Stochastic Systems (94%), Statistical Learning (87%)
- 2-year part-time study programme; completed whilst working full-time

BSc. (Hons) Economics

University of Nottingham

1st Class Honours

SKILLSET

Python | SQL | R | PySpark | Snowflake | Azure | AWS | Docker | Airflow | MLflow | PowerBI | LLMs | NLP | RAG | Deep Learning | PyTorch | Clustering | Time Series | Survival Analysis | Health Economics | Epidemiology | Mixed-effects Models | Bayesian Methods | Markov Models

PUBLICATIONS

Journal Articles

- M. Kendall et al., "Drivers of epidemic dynamics in real time from daily digital COVID-19 measurements," *Science*, vol. 385, 6710 2024. [Online]. Available: https://www.science.org/doi/10.1126/science.adm8103
- L. Ferretti et al., "Digital measurement of SARS-CoV-2 transmission risk for precision epidemiology," *Nature*, 2023. [Online]. Available: https://rdcu.be/dvLyt
- M. Kendall et al., "Epidemiological impacts of the NHS COVID-19 app in England and Wales throughout its first year," *Nature Communications*, vol. 14, 2023. [Online]. Available: https://rdcu.be/c6yas

ACHIEVEMENTS & INTERESTS

Computer Vision
Spearheaded a hackathon project focused on automating the extraction of tables and handwritten text from a backlog of source documents containing disease notifications. Leveraged Tika and Tesseract for robust object recognition on images, facilitating data extraction for analysis and integration into downstream applications.

Mathematics
Completed half of the required modules (60 UK CATS) towards the Graduate Diploma in Mathematics offered by the University of London (academic direction from London School of Economics).

Economics Instructor
Led three one-hour classes in introductory macroeconomics to the Press Office in the Department for Work and Pensions, commissioned by the Government Economic Service.